

External Adversarial Audit Report

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Executive verdict

PASS_WITH_OBSERVATIONS

The previous blocker was environmental: Block F had not completed. In this local rerun, Lean completed successfully.

- 'lake exe cache get': completed successfully.
- 'lake build': completed successfully, 8058 jobs.
- 'lake env lean gate.lean': exit 0, expected axiom/signature report.

No blocking defect was found in the completed checks. The only observation is coverage-related: Block B was a li

Per-claim table

Claim	Mathematical audit	Lean column	Evidence
Theorem 1.1 / E-9	Axiom-relative; no counterexample found in completed checks	PASS: AX1+AX2 exactly in axi	
Corollary 1.2	Axiom-relative; follows Lean gate signature	PASS: AX1+AX2 exactly in axiom report	blockF_
Proposition 10.1	Internal audit rerun passed	PASS: Layer E clean	blockF_lean_formalization/results/axiom
Theorem 3.1 / E-3.1	Exact F branch scan confirmed on $p \leq 80$ grid	PASS: Layer E clean	blockC_counterexample
Theorem 4.2 / E-4.2	Exact margin scan confirmed on $p \leq 80$ grid	PASS: Layer E clean	blockC_counterexample_
Lemmas 5.1/5.2/6.1/7.1	Internal audit rerun passed after dependency install	PASS: Layer E clean	blockE_a
Appendix B / D	Static/textual scan plus Lean gate	PASS: Layer E clean	blockF_lean_formalization/results/
AX1 / AX2	Limited bibliographic plausibility check	PASS as declared axioms; no extra axioms	blockB_exter

Findings

See 'findings/FINDINGS.csv'.

Coverage statement

Block C exact grid: $p=3..80$, $q=1..2p$, $d=0..p$ for closed-form F and Theorem 4.2 margin inequality. No violations

Block D exact arithmetic: selected coefficient/threshold identities confirmed.

Block E: copied internal scripts reran successfully after installing sympy/numpy/scipy/pulp locally.

Block F: fully executed locally from 'C:\Users\jtrav\Documents\Codex\leanrun_erdos81'; logs were written in

Block B remains limited: source-page evidence was checked, not full-paper theorem-by-theorem bibliographic certi

Reproduction appendix